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**A Recommendation System Based on Predicting Student Performance
Using Machine Learning Models**

By

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A Recommendation System Based on Predicting Student Performance Using Machine Learning Models

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ABSTRACT

Student performance prediction has become increasingly important in education because lecturers often realize that students are struggling too late, usually after midterm or final exams. At that point, it becomes so hard to provide effective support to help students improve their performance or pass the course. The early prediction systems will help identify academically at-risk students sooner and allow timely intervention before their learning difficulties become more serious.

Hence, this project will focus on predicting student performance at early coursework stages using several machine learning models. Experiments were conducted under two prediction stages based on approximately 20% and 50% of the total course assessment. Both binary classification and multi-class classification settings were evaluated using Logistic Regression, Random Forest, and Support Vector Machine (SVM).

One of the major challenges in datasets about student outcomes is class imbalance, where weak students represent only a very small portion of the data compared to the other student groups. To handle it, we generate synthetic minority samples by using BorderlineSMOTE and then filter them using an epiplexity-based sample selection method to keep only the more informative synthetic samples.

The results show that prediction performance becomes better as more coursework information becomes available in later stages of the semester. In several cases, the curated dataset also improves minority-class detection, especially when identifying weak students.

CHAPTER 1

INTRODUCTION

1.1. Background

Universities and colleges now collect a large amount of student assessment data through online learning systems and digital learning platforms. This data has created new opportunities for applying machine learning techniques to analyze either student learning behavior nor support academic decision-making processes. One important application of Educational Data Mining (EDM) is student performance prediction, where machine learning models are used to identify students who may experience academic difficulties during a course [1].

In many courses, students who are struggling are usually noticed quite late, often after midterm exams or near the end of the semester. By that time, it can already be difficult for lecturers to provide enough support for students to significantly improve their performance.

As class size increases, so does the number of classes that instructors are expected to teach and students they must track. Thus, it is not practical for instructors to manually track the individual learning development of all their students. This has led researchers to focus on developing early warning systems (EWS) to recognize at-risk students earlier and support timely academic intervention [2].

Many machine learning models have been developed to predict student performance such as Logistic Regression, Random Forest, Support Vector Machine, and ensemble learning technique. Some research has shown that multi-class educational data mining contexts can benefit from using an ensemble-based approach to develop student performance prediction models [3]. Additionally, some researchers have developed recommendation-based education support systems to bridge the gap between predictive analytics and personalized learning guidance/intervention strategies [4], [5].

However, several challenges still remain in early-stage student prediction. In many previous studies, training models are using complete course results, which makes early prediction less realistic in actual learning environments. Furthermore, educational datasets are often highly imbalanced because the number of academically weak students is usually much smaller than the number of students who successfully complete the course. This imbalance can reduce the ability of machine learning models to correctly identify minority student groups.

1.2. Problem Statement

Although numerous studies have investigated student performance prediction using machine learning techniques, several methodological and practical problems still remain.

One of the common issues is that some previous predict models are often evaluated using information that directly influences the final outcome, which is usually not available during the early stages of coursework. Such practices can introduce data leakage and overestimate model performance hence make it much more difficult in real environments.

Another difficulty is that only a small amount of assessment data is available during the early stages of a course. Because of this, predicting weak students early can be much more difficult for machine learning models.

Besides, while binary classification is useful for identifying whether a student is at risk or not, multi-class classification may offer more detailed information about different student performance levels. Hence, both prediction settings would be investigated in this study.

In addition, educational datasets are often too imbalanced because the number of weak students usually represent only a small minority of the overall population. This may bias machine learning models toward majority predictions so reducing minority-class detection capability and weakening the effectiveness of early-warning systems[2], [3]. Although resampling and synthetic data generation techniques are commonly applied to address imbalance, unrestricted augmentation may introduce noisy or low-quality synthetic samples that negatively affect model generalization[6]. Therefore, developing a more selective synthetic sample curation strategy remains an important challenge in early-stage student performance prediction.

Finally, many prediction systems mainly focus on classification results without providing practical learning support for students. Although some studies have explored recommendation-based educational support approaches, applying prediction results to generate suitable academic guidance still remains a challenging task[4], [5].

Based on these challenges, this study focuses on the following research questions:

- **RQ1:** Which machine learning model achieves the best predictive performance for early-stage student performance prediction in binary classification and multi class classification?
- **RQ2:** How does prediction performance change between the 20% and 50% coursework stages?
- **RQ3:** How does multi-class classification compare with binary classification in terms of predictive performance and risk granularity?
- **RQ4:** How can predicted student performance be utilized to design a risk-adaptive academic support recommendation framework?

1.3. Research Objectives

The objective of this study is to develop and evaluate a machine learning framework for early-stage student performance prediction and risk-adaptive academic support recommendation.

This study conducts machine learning experiments under two stages 20% and 50% of the total course assessments and also explores the use of BorderlineSMOTE and epiplexity-guided synthetic sample selection to help the models learn minority class better from imbalanced educational datasets.

The main objectives are as follows:

1. Implement and compare multiple machine learning models for binary and multi-level student performance prediction.
2. Examine how prediction performance changes between the 20% and 50% coursework stages for early-warning analysis.
3. Explore the effect of BorderlineSMOTE and epiplexity-guided synthetic sample selection on imbalanced educational data.
4. Compare model performance between the original dataset and the curated synthetic dataset.
5. To design a simple rule-based academic recommendation approach based on predicted student performance levels.

1.4. Research Contributions

The two-stage evaluation design used in this study follows the format presented by Injadat et al. This study does not reproduce that setting. Two additional features are suggested that have been omitted in the previous studies: an **epiplexity-based curation method** will be implemented for all synthetic minority-class samples prior to model training. A secondary feature is a **downstream recommendation module** which changes the output of predictions to **guide academic support recommendations**. The stage design itself is not claimed to be a contribution, the contributions come from what is added inside of it and on top of it.

In addition to the prediction structure, this study examines how different Classification-settings affect early-prediction behaviour. Both **binary and multi-class** approaches were **compared on both prediction stages** with special attention paid to how each setting can identify students who are academically weak. Binary classification showed stronger weak-recalls than multi-class settings, however multi-class settings displayed performance tier-differences which could help identify when a single label (**at-risk**) is insufficient for teacher-instruction.

Another technical difference of contributions is the **epiplexity-guided** synthetic sample curation strategy. The traditional practice of using **BorderlineSMOTE**, along with retaining all synthetically generated candidates has shown potential to be problematic, because not every candidate will generate useful information to train upon. Each group of candidates will generate incremental amounts of informational complexity of the training data - a property measured via epiplexity. Candidates do not exceed this threshold will be discarded.

Finally, the prediction outputs from the best performing models will be input into a **rule-based academic support suggestion framework** that provides **unique academic supports** for each student. Rather than giving the same intervention to each student identified as academically weak, the system utilizes the probability score predicted to divide severity into **four risk-tiers (Low, Mild, Moderate, and High)**. The type and intensity of the recommend support will vary according to the risk-tier assigned and if the prediction was made at Stage 1 or Stage 2. This design was influenced by a real-world concern: there are time constraints

remaining in a course that influence the feasibility of implementing certain types of interventions. A single fixed recommendation would fail to capture these time constraints.

1.5. Scope of the Study

This study focuses on machine learning approaches for early-stage student performance prediction using structured assessment data. The analysis is limited to course-related assessment information such as quiz scores, assignments, and midterm examination results.

The experiments are conducted under two early prediction stages corresponding to approximately 20% and 50% of the total course assessments. Both binary and multi-class classification settings are evaluated in the experiments.

To handle the class imbalance problem commonly found in educational datasets, this study explores the use of BorderlineSMOTE and epiplexity-guided synthetic sample selection during model training.

In addition, the study includes a simple rule-based academic recommendation approach based on predicted student performance levels. Behavioral, demographic, psychological, and long-term intervention data are not included in this study.

1.6. Structure of thesis

Chapter 1. Introduction: including the background story, how I identify the problem, the scope and objectives of this research. This also lightly introduces the proposed solution of the stated problem.

Chapter 2. Literature Review: reviews related studies and background knowledge relevant to this research.

Chapter 3. Methodology: including the description of dataset, model development, synthetic sampling strategy, and evaluation metrics.

Chapter 4. Experimental Results: reports and discusses experimental results across models, coursework stages, and classification settings.

Chapter 5. Recommendation System: introduces the academic support recommendation framework based on model predictive results.

Chapter 6. Conclusion & Future Work: summarizes main findings, discusses limitations, and suggests possible directions for future work

CHAPTER 2

LITERATURE REVIEW

2.1. Student Performance Prediction in Educational Data Mining

Student performance prediction has become an important research topic in Educational Data Mining (EDM) and Learning Analytics. In general, the goal is to use students' assessment results to estimate their learning performance and identify students who may need extra support during a course. Many educational institutions are interested in this area because early prediction allows lecturers to provide help before students experience serious academic difficulty [1], [7], [8].

Traditional models such as Logistic Regression are still widely applied to predict student outcomes because they are simple and easy to interpret. Besides, other studies have experimented with more complex models such as Random Forest, Support Vector Machine (SVM), Decision Tree, and Neural Networks to improve prediction performance [3], [9], [10].

Several research studies demonstrated the effectiveness of machine learning in predicting academic outcomes with the use of educational databases. One example is the work by Çırak et al., who developed a predictive model for identifying students in need of support during early stage of semester. This included the use of multiple models such as Decision Tree, Naive Bayes, and Random Forest. Their results indicated that the Random Forest model had better predictive ability than many others [2].

Another study conducted by Injadat et al. evaluated the predictive potential of student performance through the examination of two points in time. These two points corresponded to roughly 20% and 50% completion of the semester. Additionally, they used multiple machine learning methods, including Logistic Regression, Random Forest, SVM, Naive Bayes, and Neural Networks, and employed a multi-class classification method to evaluate which of these models could best predict student success. Results from this study showed that the accuracy of predictions improved over time as additional assessments were completed [3].

Despite the promising results reported in previous studies, early-stage student prediction remains challenging because only limited assessment information is available during the early part of a course. In addition, educational datasets are often highly imbalanced, where academically weak students represent only a small minority of the data. These conditions can make minority-class detection more difficult for machine learning models.

2.2. Early Warning Systems in Higher Education

Early Warning Systems (EWS) are designed to identify students who may experience academic difficulties before the end of a course or semester. The purpose of these systems is to help lecturers and educational institutions provide timely support and intervention for students who are at risk of poor academic performance or course failure [2].

Educational Data Mining techniques are now commonly used in early warning systems. Machine learning models can learn patterns from assessment data and spot students with higher

academic risk levels. Several studies also found that prediction results can become more accurate after adding more coursework information collected during the semester. However, if predictions are made too late, lecturers may have less time to provide effective support or intervention for students [3].

Early-stage prediction therefore becomes an important component of educational early warning systems. Even though prediction performance during early coursework stages may be lower due to limited available data, earlier identification of academically weak students may provide lecturers with more time to offer learning guidance, revision materials, or additional academic support.

2.3. Binary vs Multi-Class Classification in Academic Risk Detection

Student performance prediction is usually studied using either binary classification or multi-class classification approaches. In binary classification, students are divided into two groups, such as “at-risk” and “not at-risk” or “Weak” and “Good”. This method is widely used in educational early warning systems because it offers a simple way to identify students who need support [2], [11].

Many previous studies used binary classification models together with machine learning techniques such as Logistic Regression, Decision Tree, Random Forest, and SVM for student performance prediction. Compared to multi-class settings, binary classification is generally easier to implement and can still produce good results when identifying academically weak students [9], [11].

However, dividing students into only two groups may not fully reflect the differences in student performance levels. In practice, some students may fall somewhere between strong and weak academic performance. Therefore, several researchers have explored multi-class classification approaches, where students are separated into multiple categories such as Weak, Fair, and Good.

Injadat et al. studied multi-class student performance prediction using several machine learning algorithms and found that this approach could provide more detailed information about student academic performance at different coursework stages. Therefore, multi-class classification is usually more challenging because the model must distinguish between several student categories [3].

Overall, both binary and multi-class classification approaches are useful for educational prediction tasks. Binary classification is helpful for detecting students at academic risk early, while multi-class classification can provide a more detailed view of different student performance levels.

2.4. Handling Imbalanced Educational Data

In educational datasets, the imbalance is a really common issue because weak-performing students usually make up only a small part of the whole student population. Several previous studies have mentioned the effect of imbalance problems on educational prediction

models. Thai-Nghe et al. pointed out that student performance datasets often have uneven class distributions between passing and failing students, which can make prediction tasks harder for machine learning algorithms [4].

To reduce the imbalance problem, extra minority-class samples were generated using BorderlineSMOTE. Instead of randomly copying existing samples, this method focuses more on the difficult minority cases near the class boundaries, where wrong predictions happen more easily [6].

However, synthetic oversampling also has some limitations. In some cases, the generated samples may contain noise or low-quality information that can negatively affect model performance. Synthetic samples may not always help the model learn effectively, and too much oversampling can lead to reducing the model's ability to generalize well on unseen data [12].

Because of the limitation, recent studies have started exploring ways to evaluate the quality of synthetic data before using it for training. In this study, an epiplexity-guided synthetic sample selection approach is explored to remove less useful generated samples while keeping the more informative ones for training machine learning models on imbalanced educational datasets.

2.5. Recommendation Systems in Educational Contexts

Recommendation systems are commonly used in many online platforms. In education, recommendation approaches have also been explored to support student learning and provide additional academic guidance [4], [5].

Educational recommendation systems can be used to suggest learning materials, revision resources, or study support based on student academic performance and learning behavior. Several previous studies combined Educational Data Mining (EDM) techniques with recommendation methods to support personalized learning assistance.

Nguyen Thai-Nghe et al. explored the use of recommender system techniques for predicting student performance and supporting educational decision-making. Their study suggested that recommendation-based approaches may help provide additional support for students based on predicted academic outcomes [4].

Although recommendation approaches in educational environments continue to receive attention, many prediction systems still mainly focus on classification performance without providing practical learning guidance based on prediction outputs. Therefore, simple recommendation strategies based on predicted student performance levels may still provide useful academic support for educational early warning systems.

2.6. Research Gap

Based on the reviewed studies, several problems still remain in early-stage student performance prediction.

Many studies evaluate prediction models using assessment information collected during the later stages of a course, including assessments conducted close to the final examination period. This may not fully reflect realistic early prediction situations where only limited assessment data is available during the first part of the semester.

Another challenge comes from class imbalance in educational datasets, where weak students usually represent only a small minority of the data. Although synthetic sampling methods are commonly used to reduce this problem, generated synthetic samples may also introduce noisy data that negatively affects model learning.

In addition, binary and multi-class classification settings may produce different prediction behavior under early-stage learning conditions. Understanding how different models perform under these settings is important for educational early warning systems.

Finally, many prediction systems mainly focus on classification performance without providing practical learning support based on prediction results.

CHAPTER 3

METHODOLOGY

3.1. Problem Formulation

Early identification of academically at-risk students is an important task in educational data mining because it helps teachers provide timely support and reduce course failure rates. Machine learning techniques have increasingly been applied to student performance prediction using academic assessment data collected during a course [3], [9].

In this study, student performance prediction is formulated as a supervised classification problem. Given a set of assessment features collected from student coursework activities, the objective is to train machine learning models to predict the final academic performance category of each student.

Formally, let $\mathbf{x} \in \mathbb{R}^d$ denote a feature vector representing a student's academic assessment attributes, and let $y \in \mathcal{Y}$ denote the target performance label. The goal is to learn a classification function:

$$f : \mathbb{R}^d \rightarrow \mathcal{Y}$$

such that:

$$\hat{y} = f(\mathbf{x})$$

Two classification settings are investigated:

1. **Binary classification**, where students are categorized as either non-risk or at-risk.
2. **Multi-class classification**, where student performance is categorized into multiple academic levels.

These 2 settings compare prediction performance under different classification conditions and evaluate how prediction behavior changes across different student performance categories.

To simulate real world early warning conditions, the models are trained using only the assessment data available at different stages of the course. The experiments then compare how prediction performance changes between the 20% and 50% coursework stages.

The prediction outputs are later used to generate simple academic recommendations for different student groups.

3.2. Dataset Description

3.2.1. Dataset Sources

Two student performance datasets are used in this study. The data was collected from a second-year undergraduate science course at a North American university and contains academic records from **486 students**. The available features include quiz scores, assignment results, midterm examinations, and final examination outcomes. The main difference between the two datasets is their target performance categories [3], [9]:

- **Binary Dataset**, with two labels: Good vs. Weak.
- **Multi-class Dataset**, with three-class classification labels: Good, Weak, Fair.

These datasets were originally introduced in EDM research to evaluate machine learning techniques for early prediction of student academic performance[3], [9].

3.2.2. Features

The data includes many assessment measures as presented in Table 1. Each score was standardized to be on a scale of 0-100 for standardization. Any missing assessment values were replaced by a value of 0. The decimal portion of all scores was rounded up or down to the nearest integer value. The original total course grade point average was based on a maximum possible total score of 110 since there was an extra ten percent bonus that was given for Assignment 03 [3], [9].

Table 3.1: Dataset Feature Description

Feature	Description	Type
Student ID	Student identifier	Nominal
Quiz01	Quiz 1 score	Numeric
Assign.01	Assignment 1 score	Numeric
Midterm	Midterm examination score	Numeric
Assign.02	Assignment 2 score	Numeric
Assign.03	Assignment 3 score	Numeric
Final Exam	Final examination score	Numeric

Feature	Description	Type
Final Grade	Final course mark	Numeric
Student Category	Performance category	Nominal

3.2.3. Target Variables

The dataset contains categorical labels representing student academic performance under two classifications: binary and multi-class classification.

Binary Classification

In the binary classification, students are divided into two groups:

- **Good (G):** $60 \leq \text{Course grade} \leq 100$
- **Weak (W):** $\text{Course grade} \leq 60$

The 60% threshold is commonly treated as the passing grade in undergraduate courses. Students in the Weak category are therefore considered academically at risk [5].

Multi-Class Classification

In the multi-class setting, student performance is divided into three categories:

- **Good (G):** $70 \leq \text{Course grade} \leq 100$
- **Fair (F):** $51 \leq \text{Course grade} \leq 69$
- **Weak (W):** $\text{Course grade} \leq 50$

Compared to binary classification, this setting provides more detailed performance levels by separating moderate-performing students from both strong and academically weak students.

3.3. Early-Stage Feature Design

3.3.1. Stage 1: Early Coursework (20%)

This stage represents the earliest prediction stage. The selected features correspond to assessment components whose combined weight is close to 20% of the total coursework. The values in brackets represent the percentage weights of the assessment components in the overall course grade:

- **Quiz01 [10]**
- **Assignment01 [8]**

As can be seen, there were only two features selected leading to the lack of information, so the prediction task would be more challenging and relies on limited indicators of student performance.

This experiment examines whether early coursework results can already provide enough information for predicting final student performance. Making predictions at an earlier stage may give lecturers more time to identify struggling students and provide support before their academic situation becomes more difficult later in the semester.

3.3.2. Stage 2: Mid-Course Assessment (50%)

With stage 2, the combined weight of course assessments is about 50% coursework. Therefore, this stage includes more student performance information.

- **Quiz01 [10]**
- **Assignment01 [8]**
- **Midterm [20]**
- **Assignment02 [12]**

Compared to Stage 1, this stage contains more assessment information about student academic performance. With additional features available, the models are expected to generate more stable and reliable prediction results.

Comparing the results between Stage 1 and Stage 2 also helps show how prediction performance changes when more coursework information becomes available during the semester.

3.3.3. Data Leakage Prevention

During the feature selection process, only assessment information available before each prediction stage was used for model training. Features collected near the end of the course were removed from the training data because they already contain too much information about the final course result.

The following were not included as input features:

- **Final Exam**
- **Final Grade**

These variables come from the later part of the course so they already reveal a lot about the final course result. Including them in the training process could make the prediction task unrealistically easier because the models would indirectly access information strongly connected to the target labels.

In reality, lecturers need to identify weak students earlier in the semester, before final assessments and late-course assignments are completed. Because of this, the experiments in this study only use assessment information that would realistically be available during the corresponding prediction stage.

3.4. Data Preprocessing

3.4.1. Exploratory Data Analysis

Several exploratory analyses were performed before model training in order to better understand the dataset characteristics and class distribution:

1. **Missing value and duplicate checking:** no missing value and duplicate
2. **Class distribution analysis:**

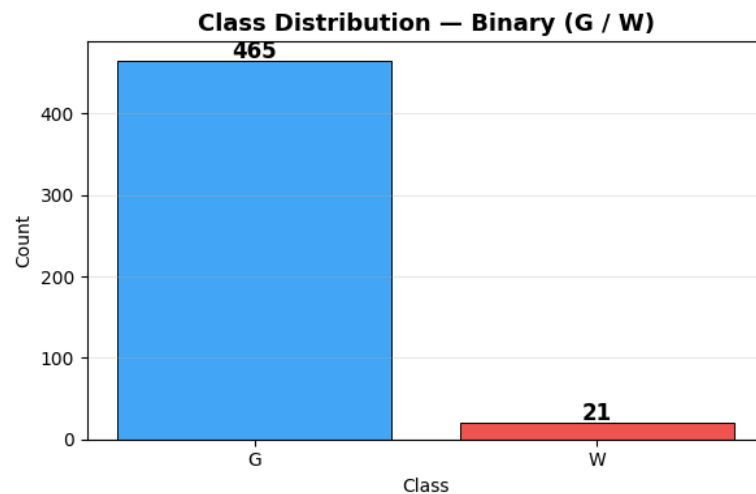


Figure 3.1: Binary Dataset - Class Distribution

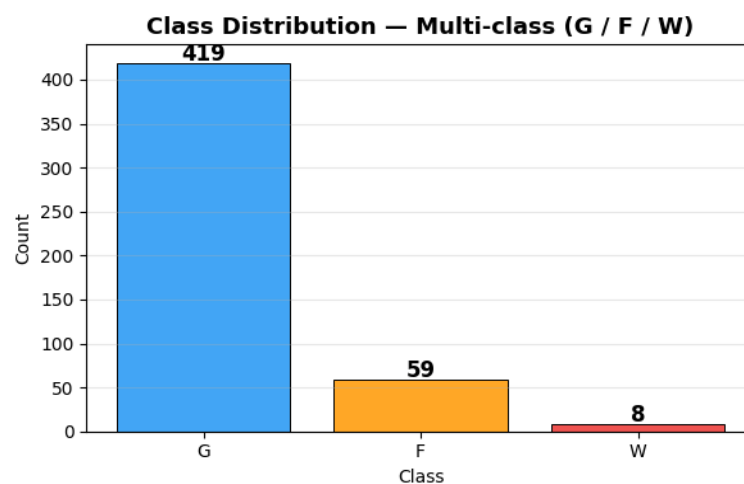


Figure 3.2: Multi-class Dataset - Class Distribution

From Figure 1, the binary dataset contains only 21 Weak students compared to 465 students in the Good category. Similarly in Fig 2, the multi-class dataset only has 8 Weak students, while Good and Fair students are 419 and 59, respectively.

Both figures show highly imbalanced and heavily underrepresented data in minority class. This makes the prediction task more difficult, as machine learning models may become biased toward the larger classes and struggle to identify weak students correctly.

3. Feature distribution analysis: Histogram visualization for all features

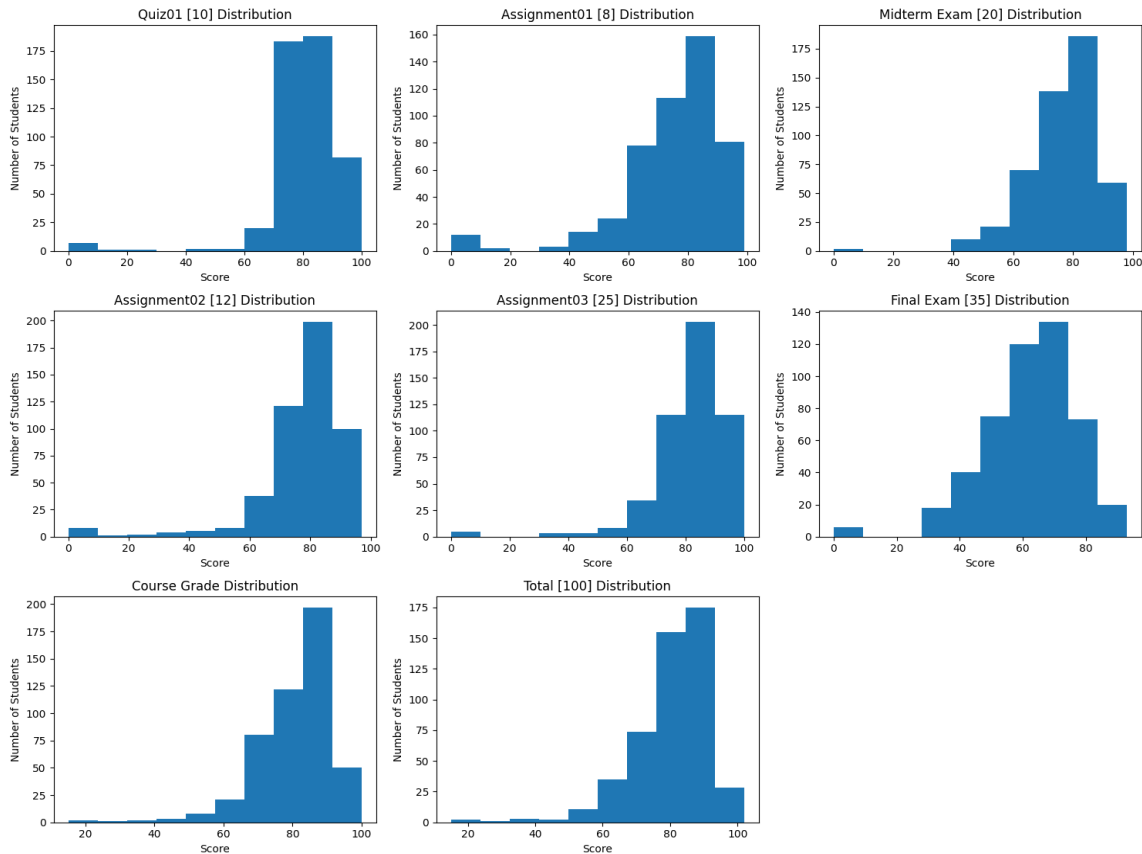


Figure 3.3 : Feature Distributions

Most scores were distributed around the higher score range, especially for assignments and quizzes. A large number of students achieved scores between around 70 and 90. Assignment02, Assignment03, Course Grade, and Total score distributions were more concentrated compared to the other assessments.

The Final Exam scores looked less concentrated than the other assessments. More students received lower scores in the final exam compared to quizzes and assignments. A few low-score cases could also be seen in several features, but these students only represented a small part of the dataset.

4. Outlier checking:

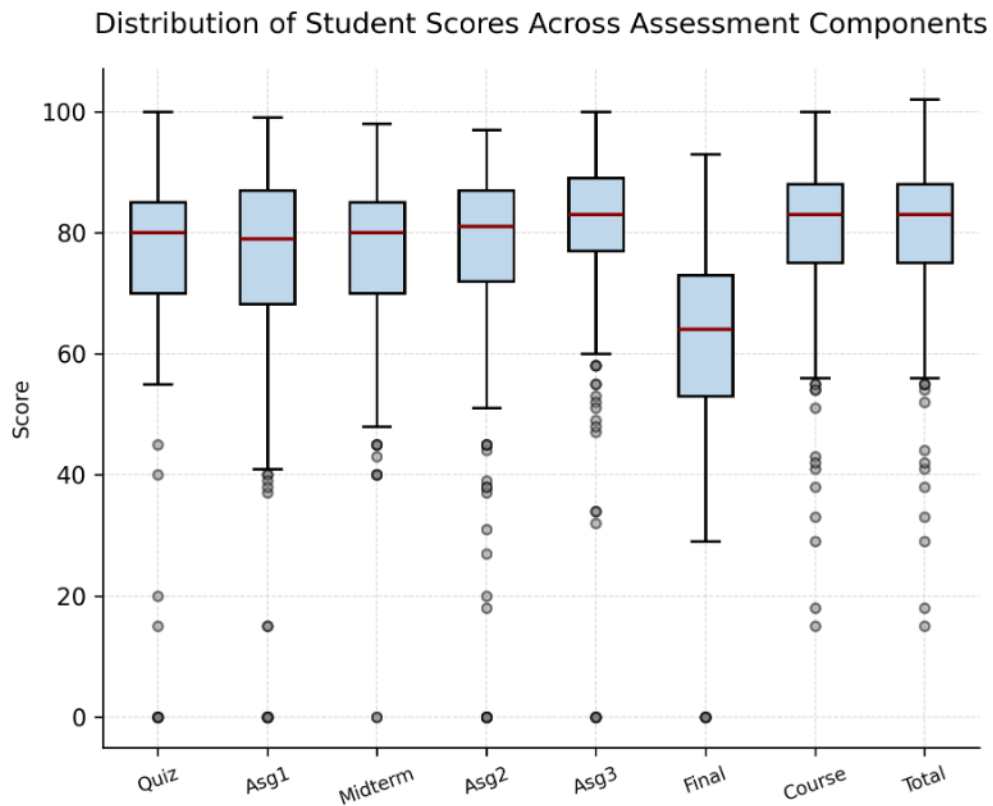


Figure 3.4: Distribution of Student Scores Across Assessment Components

Based on the boxplot above, most components have relatively high scores. The median values for quizzes, assignments, and the midterm remained fairly high, while the score ranges were not extremely wide.

Compared to the other assessments, the Final Exam had a wider spread and lower median score. Several lower-score outliers were also visible across different assessment components. These cases mainly belonged to a smaller group of weaker-performing students in the dataset.

5. **Correlation Analysis:** Heatmap was generated to examine relationships between features.

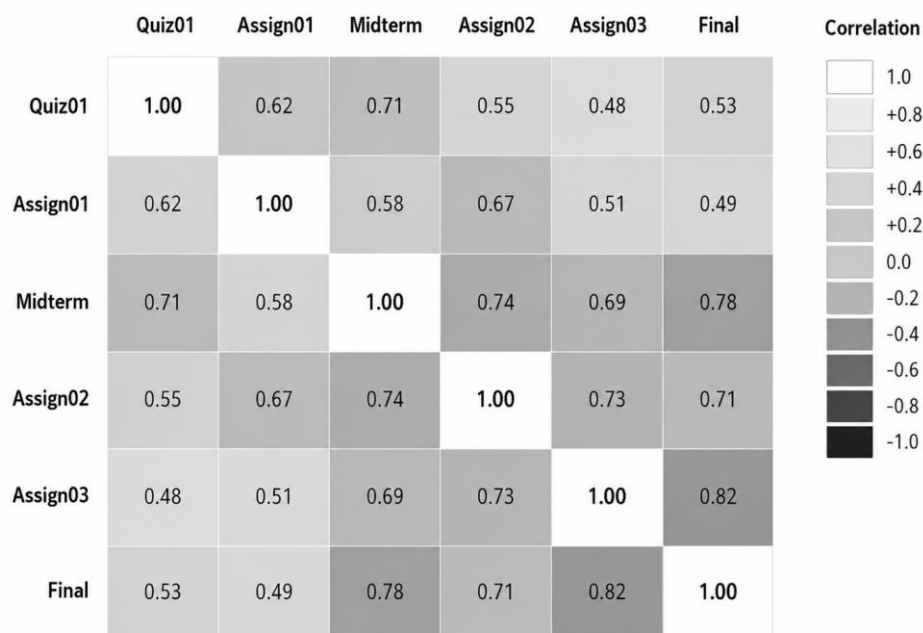


Figure 3.5: Correlation Heatmap

3.4.2. Label Encoding

Target labels were encoded into numerical for machine learning compatibility. Class = W was assigned as 1 for both datasets to maintain consistency and simplified the implementation.

Binary Dataset:

```
# Encoded target Labels (binary dataset)
binary_le = LabelEncoder()
binary_df["Class"] = binary_le.fit_transform(binary_df["Class"])
print("Encoded Labels (binary dataset):", binary_le.classes_)
```

```
Encoded Labels (binary dataset): ['G' 'W']
```

Multi-class Dataset:


```
# Encoded Labels(multiclass dataset)
mapping = {
    "G": 0,
    "W": 1,
    "F": 2
}
multi_df["Class"] = multi_df["Class"].map(mapping)
```

3.4.3. Data Splitting

The datasets were divided into 70% training data and 30% testing data. The models were trained using the training set and later evaluated using the testing set.

A stratified train-test split was applied so that the class distribution remained similar in both the train and test sets. This was important because the datasets contain a relatively small number of weak students compared to the other performance groups.

3.5. Synthetic Sample Generation

To increase the number of minority-class samples, synthetic data was generated after splitting the dataset into training and testing sets. Only the training set was augmented, while the testing set remained unchanged. This generation only used student grades collected before the final exam because they do not have too much data about the course result.

BorderlineSMOTE was applied to generate additional minority-class samples. Compared with simple random oversampling, BorderlineSMOTE pays more attention to minority samples located near the class boundaries, where classification is usually more difficult [6].

Since the number of weak students was very small, most generated samples were added to the Weak class. For the multi-class dataset, some additional samples were also generated for the Fair class.

During the experiments, a few generated records looked very different from the original student scores. This was easier to notice in cases where the minority class only had a limited number of samples. In some cases, using all generated samples together made the predictions become worse [6], [13].

The synthetic sample generation pipeline:

1. Split the original dataset into train-test.
2. Identify minority class samples from the training data.
3. Generate synthetic candidate samples using BorderlineSMOTE.
4. Evaluate generated samples using the proposed epiplexity-guided selection process.
5. Retain selected synthetic samples for the final model.

This process attempts to improve minority-class learning while reducing the influence of low-quality synthetic samples during training.

3.6. Epiplexity-Based Sample Evaluation

Finzi et al. introduced the idea of epiplexity to describe how much useful learning information exists inside a dataset [12]. In general, data with clearer patterns is usually easier for machine learning models to learn from.

In this work, epiplexity was adapted to filter generated synthetic samples before the final training step. Instead of applying the full theoretical definition, a simpler loss-based version was applied during training.

The epiplexity score was calculated from the difference between the loss values during training and the final converged loss.[12]:

$$\text{Epiplexity} = \sum_{i=1}^n (L_i - L_{final})$$

where:

- L_i is the loss at training epoch i
- L_{final} is the final loss after convergence

Logistic Regression was used here as prone models because the loss values are easier to follow during training. The goal of this step was simply to see whether the generated samples changed the learning process in a positive or negative way [14].

During the experiments, the generated samples were not used all at once. They were checked first before being added into the training dataset. After adding each group into the training data, the epiplexity score was recalculated. Samples were kept only if the score increased enough compared to the previous result.

After this step, the remaining synthetic samples were combined with the original training data to build the final dataset used in the experiments. This was mainly done to reduce some noisy generated samples before the final training stage.

3.7. Model Development

3.7.1. Model Definition

The **Logistic Regression, Random Forest, and Support Vector Machine (SVM)** were implemented for evaluating the effectiveness of machine learning approaches in early warning student performance prediction. These models were selected because they represent different machine learning paradigms and have been widely used in classification problems, including educational data mining tasks [9], [11].

```

MODELS = {
    "LR": LogisticRegression(
        max_iter=2000,
        solver='liblinear',
        class_weight='balanced',
        random_state=RANDOM_STATE
    ),
    "RF": RandomForestClassifier(
        n_estimators=200,
        class_weight='balanced',
        random_state=RANDOM_STATE
    ),
    "SVM": SVC(
        kernel='rbf',
        probability=True,
        class_weight='balanced',
        random_state=RANDOM_STATE
    )
}

```

Figure 3.6: Model definition

The parameters settings of these 3 models were designed to improve learning stability and minority class sensitivity at an imbalanced dataset. Since the number of weak students was much smaller than others, `class_weight = 'balanced'` was applied to help these models reduce majority class bias which can enhance the identification of academically at-risk students. A `random_state = 42` was also used to ensure experimental reproducibility across different training runs. The same set of models was applied to both **binary classification** and **multi-class classification** settings to ensure a consistent and fair comparison in all experiments.

3.7.2. Model Training Strategy

The experiments investigated model performance at different stages of the semester using 2 early prediction stage:

- **Stage 1 - 20% Coursework**
- **Stage 2 - 50% Coursework**

For each stage, models were evaluated using 2 different dataset settings:

- Original dataset
- Curated dataset produced by using the proposed epiplexity-based synthetic sample curation framework

The curated dataset was created through 3 steps:

1. Generating synthetic minority samples using BorderlineSMOTE
2. Epiplexity-based synthetic sample evaluation
3. Selecting suitable synthetic samples to build the final curated dataset

Multiple machine learning algorithms were then trained and tested on both dataset settings. This setup makes it possible to compare:

- Performance differences between machine learning models
- The effectiveness of the proposed imbalance handling approach
- Prediction results at different coursework stages
- Binary and multi-class classification outcomes

This analysis provides a clearer understanding of whether curated synthetic sampling can improve the early detection of at-risk learners.

3.8. Model Evaluation and Selection

The experiments were conducted under both binary and multi-class classification. Model performance evaluation using:

- **Recall_W:** Proportion of actual Weak students correctly identified
- **Macro-F1 Score:** Balanced performance of different classes
- **Confusion Matrix:** Detailed classification breakdown
- **Accuracy:** Overall correct predictions

Although accuracy is still useful for measuring general classification performance, relying only on accuracy may be misleading for imbalanced educational datasets. A model may still produce very high accuracy even when several weak students are predicted incorrectly, since weak students only account for a small portion of the dataset.

For this reason, Macro-F1 score and Weak Recall were given additional attention during the experimental analysis. Therefore, these ones used to observe how well the models handled weak-student prediction across the experiments.

CHAPTER 4

EXPERIMENTAL RESULTS

4.1. Overview

This chapter presents the experimental results of the proposed student performance prediction framework under both binary and multi-class classification settings. The experiments were conducted across two coursework stages (20% and 50% completion) to evaluate the effectiveness of early-warning prediction.

Results are organized based on the research questions and include comparative analysis across multiple machine learning models, dataset configurations, and classification settings. Accuracy, Recall, Macro-F1 score, and confusion matrices are utilized to assess overall classification performance and minority-class detection capability.

4.2. Binary Classification Results

This section the experimental results for the binary classification setting, where students were divided into two groups: Good (G) and Weak (W).

4.2.1. Train- test split

The dataset was divided using stratified sampling. Example output:

```
Train: Counter({np.int64(0): 325, np.int64(1): 15})
Test : Counter({np.int64(0): 140, np.int64(1): 6})
```

Figure 4.1: Binary Dataset Distribution in train-test sets

The severe imbalance problem can be observed from the minority-class distribution.

4.2.2. Generate Synthetic

Synthetic minority samples were generated using BorderlineSMOTE before model training. Example output:

```
=====
STEP 1 – GENERATE SYNTHETIC CANDIDATES
=====

Synthetic samples generated:
33

Synthetic class distribution:
Counter({np.int64(1): 33})
```

Figure 4.2: Synthetic Candidate Generation Result (Binary dataset)

The generated synthetic samples were treated as candidate samples and evaluated before being added to the training dataset.

4.2.3. Synthetic Sample Evaluation

The synthetic samples were evaluated using the proposed epilexity-based filtering mechanism. Example output:

```
=====
STEP 2 – EVALUATING SYNTHETIC SAMPLES
=====

Baseline Epilexity: 0.405293

-----
Chunk #1
Samples: 5
Old Epilexity : 0.405293
New Epilexity : 0.244853
Gain          : -0.160439
Decision      : REJECT

-----
Chunk #2
Samples: 5
Old Epilexity : 0.405293
New Epilexity : 0.268359
Gain          : -0.136934
Decision      : REJECT
```

Figure 4.3: Example Output of the Epilexity-Based Filtering Process

Only chunks that improved epilexity beyond the predefined threshold were retained.

```
=====
SYNTHETIC SAMPLE EVALUATION RESULTS
=====
```

	chunk	samples	new_epilexity	gain	decision
0	1	5	0.244853	-0.160439	REJECT
1	2	5	0.268359	-0.136934	REJECT
2	3	5	0.136415	-0.268878	REJECT
3	4	5	0.183119	-0.222174	REJECT
4	5	5	0.443656	0.038364	KEEP
5	6	5	0.410595	-0.033061	REJECT
6	7	3	0.647065	0.203409	KEEP

Figure 4.4: Synthetic sample results (Binary dataset)

To evaluate the effectiveness of the proposed synthetic sample generation and epilexity-inspired filtering framework, machine learning models were trained and tested on both the original dataset and the curated augmented dataset.

```
=====
FINAL COMPARISON
=====
```

	Dataset	Model	Macro_F1	Accuracy	Weak_Recall
0	Original	LR	0.867754	0.972603	1.000000
1	Original	RF	0.896454	0.986301	0.666667
2	Original	SVM	0.824308	0.965753	0.833333
3	Curated	LR	0.894585	0.979452	1.000000
4	Curated	RF	0.896454	0.986301	0.666667
5	Curated	SVM	0.959746	0.993151	1.000000

Figure 4.5: Evaluation Results for the Synthetic Sample Filtering Process (Binary dataset)

The figure shows that the curated dataset did not improve all models consistently. Logistic Regression and Random Forest produced only minor differences between the original and curated datasets. However, SVM showed more noticeable improvement.

Since SVM is generally more sensitive to noisy samples and decision-boundary changes, the improvement suggests that the proposed filtering process may have removed some low-quality synthetic samples from the augmented dataset. Although the improvement was not consistent across every model, the curated dataset was still retained for the later experiments in order to further evaluate its behavior.

4.2.4. Curated Dataset Construction

After evaluation, the accepted synthetic samples were merged with the original training dataset. Example output:

```
=====
STEP 3 - BUILD CURATED DATASET
=====

Original Distribution:
Counter({np.int64(0): 325, np.int64(1): 15})

Selected Synthetic Samples:
8

Final Curated Distribution:
Counter({np.int64(0): 325, np.int64(1): 23})
```

Figure 4.6: Class Distribution After the Filtering Process (Binary dataset)

4.2.5. Stage 1 (20% Coursework)

=====					
FINAL COMPARISON					
=====					
	Dataset	Model	Macro_F1	Accuracy	Weak_Recall
0	Original	LR	0.636816	0.890411	0.666667
1	Original	RF	0.739286	0.958904	0.500000
2	Original	SVM	0.655892	0.904110	0.666667
3	Curated	LR	0.620012	0.876712	0.666667
4	Curated	RF	0.739286	0.958904	0.500000
5	Curated	SVM	0.677941	0.917808	0.666667

Figure 4.7: Binary Classification - Stage 1 result

Discussion:

The curated dataset showed improvements in weak-student detection compared to the original dataset.

After the filtering step, Random Forest have the overall scores and Macro-F1 dropped a little, while the Weak Recall is unchanged after using the curated data. This suggests that the model still failed to detect the same weak student cases as before, but some additional Good students may have been predicted incorrectly as Weak after the filtering step. Random Forest results stayed almost the same. One possible reason is that the Stage 1 dataset only contained a very small number of features, so the generated synthetic samples did not create enough changes to affect the linear decision boundary of the model.

On the other hand, the filtered synthetic samples seemed to help SVM more than the other models during training. Since SVM is usually more affected by small changes in the dataset, the filtering step may have removed some noisy generated samples before the final experiments. Its Accuracy increased from 0.904 to 0.917 and the Macro-F1 score also became slightly higher.

As can be seen in curated dataset results, Random Forest achieved stronger overall scores, but its weak-student detection at 0.5 which is lower than SVM. Logistic Regression detected weak students same as SVM, although its overall performance was not as strong. Based on the overall results after filtering, SVM was chosen for this case.

4.2.6. Stage 2 (50% Coursework)

=====					
FINAL COMPARISON					
=====					
	Dataset	Model	Macro_F1	Accuracy	Weak_Recall
0	Original	LR	0.824308	0.965753	0.833333
1	Original	RF	0.896454	0.986301	0.666667
2	Original	SVM	0.785294	0.945205	1.000000
3	Curated	LR	0.843850	0.965753	1.000000
4	Curated	RF	0.896454	0.986301	0.666667
5	Curated	SVM	0.802969	0.952055	1.000000

Figure 4.8: Binary Classification - Stage 2 result

Discussion:

The Logistic Regression results increased slightly after the filtering process. In this case, the model detected all weak students in the testing set. Macro-F1 also improved compared to the earlier results.

Random Forest continued to produce the strongest overall scores. Both Accuracy and Macro-F1 became higher after training on the curated dataset, while Weak Recall remain unchanged.

SVM still detected all weak students correctly on both datasets, but its overall score is the lowest. However, the curated dataset seemed to influence SVM more clearly than the other models. After the filtering process, both Accuracy and Macro-F1 increased slightly while Weak Recall remained at 1.

This may suggest that removing some noisy synthetic samples helped make the training data more stable for SVM. Since SVM is usually more sensitive to small data changes, the effect of the filtering process became easier to observe through its results.

The best model in the Stage 2 binary classification experiment is Logistic Regression on the curated dataset. Random Forest had the highest Accuracy, but Logistic Regression successfully detected all weak students and still maintained a high Accuracy of 0.973. In real education environment, early-warning systems are usually expected to prioritize the detection of struggling students, even if a small number of good ones are incorrectly classified as weak.

4.3. Multi-class Classification Results

This section examines the results of the multi-class prediction models across three student categories: Good (G), Fair (F), and Weak (W). The experiments aim to explore how well the models can capture differences in student performance levels while still effectively recognizing students who may require early academic support.

4.3.1. Train-test split

The dataset was divided using stratified sampling.

Example output:

```
Train: Counter({np.int64(0): 293, np.int64(2): 41, np.int64(1): 6})
Test : Counter({np.int64(0): 126, np.int64(2): 18, np.int64(1): 2})
```

Figure 4.9: Multi-class Dataset Distribution in train-test sets

The severe imbalance problem can be observed from the minority-class distribution.

It should also be noted that the testing set only contained two weak students in this experiment. Because of this, predicting only one weak student incorrectly would immediately reduce the Weak Recall score to 0.5. As a result, the Weak Recall values in this setting were highly sensitive to small prediction changes.

4.3.2. Generate Synthetic

Synthetic minority samples were generated using BorderlineSMOTE before model training. Example output:

```
=====
STEP 1 – GENERATE SYNTHETIC CANDIDATES
=====

Synthetic samples generated:
43

Synthetic class distribution:
Counter({np.int64(2): 24, np.int64(1): 19})
```

Figure 4.10: Synthetic Candidate Generation Result (Multi-class dataset)

The generated synthetic samples were treated as candidate samples and evaluated before being added to the training dataset.

4.3.3. Synthetic Sample Evaluation

The synthetic samples were evaluated using the proposed epiplexity-based filtering mechanism. Example output:

```
=====
STEP 2 - EVALUATING SYNTHETIC SAMPLES
=====

Baseline Epiplexity: 0.540581

-----
Chunk #1
Samples: 5
Old Epiplexity : 0.540581
New Epiplexity : 0.526864
Gain           : -0.013717
Decision       : REJECT

-----
Chunk #2
Samples: 5
Old Epiplexity : 0.554135
New Epiplexity : 0.554135
Gain           : 0.013555
Decision       : KEEP
```

Figure 4.11: Example Output of the Epiplexity-Based Filtering Process (Multi-class dataset)

Only chunks that improved epiplexity beyond the predefined threshold were retained.

```
=====
SYNTHETIC SAMPLE EVALUATION RESULTS
=====
```

	chunk	samples	new_epiplexity	gain	decision
0	1	5	1.544693	-3.206489	REJECT
1	2	5	1.274748	-3.476434	REJECT
2	3	5	1.765191	-2.985991	REJECT
3	4	5	1.393468	-3.357714	REJECT
4	5	5	4.182545	-0.568637	REJECT
5	6	5	5.481203	0.730021	KEEP
6	7	5	3.301042	-2.180161	REJECT
7	8	5	3.207848	-2.273355	REJECT
8	9	3	9.753594	4.272391	KEEP

Figure 4.12: Synthetic sample results (Multi-class dataset)

Selected models were applied on both the original dataset and the curated augmented dataset to examine the effect of the proposed synthetic sample generation and filtering approach.

```
=====
FINAL COMPARISON
=====
```

	Dataset	Model	Macro_F1	Accuracy	Weak_Recall
0	Original	LR	0.676687	0.904110	0.5
1	Original	RF	0.678390	0.924658	0.5
2	Original	SVM	0.757031	0.890411	0.5
3	Curated	LR	0.734641	0.904110	0.5
4	Curated	RF	0.730159	0.931507	0.5
5	Curated	SVM	0.757031	0.890411	0.5

Figure 4.13: Evaluation Results for the Synthetic Sample Filtering Process (Multi-class dataset)

This shows that the curated dataset did not produce the same effect across all models. Logistic Regression and SVM performances are still stable after augmentation, whereas Random Forest improved a little in terms of Accuracy and Macro-F1 score.

The changes were fairly small, but the curated dataset still showed some potential benefits in certain cases. For that reason, it was included in the later experiments for additional evaluation.

4.3.4. Curated Dataset Construction

After evaluation, the accepted synthetic samples were merged with the original training dataset. Example output:

```
=====
STEP 3 – BUILD CURATED DATASET
=====

Original Distribution:
Counter({np.int64(0): 293, np.int64(2): 41, np.int64(1): 6})

Selected Synthetic Samples:
8

Final Curated Distribution:
Counter({np.int64(0): 293, np.int64(2): 49, np.int64(1): 6})
```

Figure 4.14: Class Distribution After the Filtering Process (Multi-class dataset)

4.3.5. Stage 1 (20% Coursework)

=====					
FINAL COMPARISON					
=====					
	Dataset	Model	Macro_F1	Accuracy	Weak_Recall
0	Original	LR	0.563057	0.869863	0.5
1	Original	RF	0.448066	0.780822	0.5
2	Original	SVM	0.511441	0.678082	0.5
3	Curated	LR	0.610303	0.863014	0.5
4	Curated	RF	0.467725	0.787671	0.5
5	Curated	SVM	0.529055	0.712329	0.5

Figure 4.15: Multi-class Classification - Stage 1 result

Discussion – stage 1:

For this Stage 1 setting, the results between the original and curated datasets did not change much. Logistic Regression performance went down a little after the filtering step. Random Forest increased slightly in Accuracy. SVM changed more noticeably compared to the other models after using the curated dataset. The Accuracy increased from 0.68 to 0.71, and the Macro-F1 score became slightly higher than before. In this experiment, Weak Recall stayed the same across all three models.

From these results, the filtering process did not create major changes in weak-student detection during the Stage 1 setting. However, the curated dataset still appeared to help SVM and Logistic Regression learn better compared to the original dataset. Since Stage 1 only used a very small number of assessment features, the prediction task at this point was still quite difficult for all models.

Overall, Logistic Regression trained on the original dataset produced the best results for the Stage 1 multi-class classification setting.

4.3.6. Stage 2 (50% Coursework)

=====					
FINAL COMPARISON					
=====					
	Dataset	Model	Macro_F1	Accuracy	Weak_Recall
0	Original	LR	0.676687	0.904110	0.5
1	Original	RF	0.623085	0.897260	0.5
2	Original	SVM	0.488329	0.842466	0.0
3	Curated	LR	0.734641	0.904110	0.5
4	Curated	RF	0.687641	0.910959	0.5
5	Curated	SVM	0.669444	0.863014	0.5

Figure 4.16: Multi-class Classification - Stage 2 result

Discussion - stage 2

For this experiment, Logistic Regression produced identical results on both the original and curated datasets. The filtering process did not change the model performance in this case, with Macro-F1 grew to 0.73 and Weak Recall staying at 0.5.

Random Forest showed a small improvement after using the curated dataset. Both Accuracy and Macro-F1 became gradually higher, although the Weak Recall score still remained unchanged.

Among the three models, SVM showed the largest change after training on the curated dataset. The original SVM model was unable to detect weak students correctly, resulting in a Weak Recall of 0.0. After using the curated dataset, Weak Recall increased to 0.5, while Macro-F1 and Accuracy also became higher compared to the original result.

From these results, the filtering step appeared to help SVM more clearly than the other models in this experiment. The curated dataset improved weak-student detection for SVM and also increased the overall classification scores after training.

Logistic Regression produced the strongest overall results in this experiment. Its Macro-F1 score was the highest among all models, while Accuracy and Weak Recall remained comparable to the other approaches.

4.4. Discussion Research Question

4.4.1. RQ1: Best Model for Early-Stage Prediction

The best model changed depending on the experiment setting.

For the Stage 1 binary classification results, SVM on the curated dataset was selected because the filtering step improved Weak Recall compared to the other models. Random Forest still had higher overall scores, but weak-student detection was lower than SVM in this case. In the Stage 2 binary experiments, Logistic Regression on the curated dataset gave the best suitable. Both Accuracy and Macro-F1 became very high after the filtering process, while Weak Recall peak at 1.0.

In the multi-class setting, Logistic Regression performed more consistently than the other models across both stages. The results stayed higher than Random Forest and SVM in most cases, especially for Macro-F1 and Accuracy.

For the multi-class, Logistic Regression gave the strongest overall performance in both stages. In Stage 1, the Logistic Regression results stayed higher than Random Forest and SVM across most evaluation metrics. The same can also be seen in the Stage 2, where Logistic Regression still produced the highest overall scores.

4.4.2. RQ2: Performance Change Between Stages

All Stage 2 results became higher than Stage 1 for both binary and multi-class experiments. Since more coursework scores were available at this point, the models had more information during training compared to the earlier stage.

In the binary experiments, most models achieved higher Accuracy, Macro-F1, and Weak Recall during Stage 2. When more coursework scores were added in Stage 2, the weak-student results also improved in most experiments.

The same trend could still be seen in the multi-class setting. The scores became higher after adding more assessment features, even though multi-class prediction was still more difficult overall.

Implication for RQ2:

These findings suggest that the models appeared to perform more reliably when more assessment information became available later in the semester.

4.4.3. RQ3: Binary vs. Multi-class Comparison

In most experiments, the binary results were usually higher than the multi-class results. This could be seen from both Accuracy and Macro-F1 scores across different models.

The multi-class experiments were more difficult in general. The weak student was already limited, and the models still needed to distinguish several performance categories during prediction.

Although multi-class classification produced lower overall performance, it still provided more detailed information about different student performance levels. Instead of only separating students into weak and non-weak groups, the multi-class setting allowed the models to distinguish among Good, Fair, and Weak students.

Implication for RQ3:

Hence, binary classification seemed more suitable for simple early-warning prediction, while multi-class classification is more useful for detailed academic monitoring and personalized intervention planning.

4.4.4. RQ4: How Predicted Risk Levels Can Support a Risk-Adaptive Recommendation Framework

The experimental results demonstrate that student performance prediction models can be effectively integrated into a risk-adaptive academic support recommendation framework. By predicting student risk levels at different stages of the course, the proposed system can provide personalized intervention strategies based on the predicted likelihood of academic risk.

Both binary and multiclass classification approaches were tested in this study. Even though the multiclass setting can describe student performance in more detail, the binary setting showed more stable results when detecting students who may be academically at risk. This is because students in the Fair category do not always remain stable throughout the semester and can fall into the Weak group if support is not provided early enough. Hence, the binary framework was selected for the recommendation system.

Within the proposed framework, students are first classified into two categories: Good and Weak. Based on the predicted risk label, the recommendation system generates adaptive academic support strategies:

- Weak students receive early-warning notifications, additional learning materials, tutoring support, revision planning.
- Good students may receive performance maintenance recommendations, enrichment activities, and continuous progress monitoring suggestions.

To further support adaptive intervention, prediction probability scores are additionally used to estimate risk severity levels. When the predicted Weak probability is higher, the system may provide more intensive intervention strategies and additional academic guidance. Students with lower risk levels may only receive simple monitoring recommendations.

Based on the experimental results, different models worked better at different stages of the course. During the earlier prediction stage, SVM trained on the curated dataset showed

better weak-student detection performance at the 20% coursework stage. Random Forest had stronger overall results after more assessment became available during the 50% coursework stage.

4.4.5. Summary of Models for Recommendation Framework

The following models are selected as inputs for the recommendation framework:

Stage	Classification Setting	Selected Model	Selection Rationale
Stage 1 (20%)	Binary	SVM	More reliable weak-student detection during the early coursework stage
Stage 2 (50%)	Binary	Logistic Regression	Achieved more stable overall prediction performance and strong detect Weak during the later coursework stage

The selected models were chosen based on their balance between overall performance, minority-class detection, and model stability across datasets.

4.4.6. Limitations

Several limitations should be acknowledged:

Dataset Size and Imbalance.

The dataset is small and highly imbalance, especially for the minority class. Although synthetic sample generation and epiplexity-based curation were applied to mitigate this, the limited of minority samples still affect the stability and generalizability of the results. Therefore, the experimental findings should be interpreted cautiously when applied to larger or more diverse educational populations.

Single Institution and Course Context.

The experiments in this work were carried out using data from just one course at one institution. So, the results may look different if the framework is tested in other educational settings.

Feature Scope.

This study primarily relies on assessment and engagement-related features. Additional information such as attendance records, learning management system activity, demographic factors, or psychological indicators may further improve prediction performance, especially during early-stage prediction where limited course information is available.

The prediction models mainly use assessment-related and engagement-related features. Other types of information, such as attendance data, LMS activity, demographic information, or psychological factors, were not included in this study. Including these additional features may improve prediction performance, particularly during the early stages of the semester when limited coursework information is available.

Synthetic Sample Dependence.

This study also relies quite a lot on synthetic sample generation. Even though the filtering process removes some noisy synthetic samples, the generated data still comes from the original minority-class data.

CHAPTER 5

RECOMMENDATION SYSTEM

5.1. System Overview

The recommendation system was developed based on the student performance prediction framework discussed in previous chapters. After predicting student performance using machine learning models, the system generates different academic support recommendations according to the predicted student risk level.

The main goal of the system is to help identify students who may require additional academic support during different stages of the semester. Instead of only predicting student performance, the framework also provides simple recommendation strategies that may help instructors support students earlier.

The framework mainly includes:

- student performance prediction
- risk identification
- rule-based recommendation generation
- academic support recommendations

5.2. Recommendation System Architecture

The proposed recommendation system consists of four main components:

1. Student data input
2. Performance prediction
3. Recommendation generation
4. Recommendation output

The overall workflow of the recommendation system is illustrated as follows:

Student Assessment Data → Prediction Module → Risk Assessment →
Recommendation Generation → Recommendation Output

The framework consists of three main modules:

1. Prediction Module

The first part of the framework is the prediction module. This module uses the machine learning models from Chapter 4 to predict whether a student may be academically at risk based on coursework-related features.

For the binary setting, students are grouped into two categories:

- Good (G): non-risk students
- Weak (W): at-risk students

Different methods may be selected based on the time of the term. The Stage 1 binary model will most likely be used for early predictions when there is limited course work information. On the other hand, the Stage 2 model will probably be employed closer to the end of the semester as more assessment data becomes available to make more accurate predictions.

The prediction results are then passed to the risk assessment module.

2. Risk Assessment Module

The risk assessment module converts the prediction results into easier-to-understand academic risk levels.

Students predicted as Good are assigned to the “Low Risk” category. For students predicted as Weak, a single binary label is insufficient to determine the appropriate level of academic intervention.

Since not all students classified as Weak carry the same degree of risk. For example, a student whose predicted Weak probability is close to the decision boundary could possibly recover with minimal support, whereas a student with a very high Weak probability is likely to need more intensive assistance.

By only using the students' predicted class labels, we can provide no differentiation among those students at risk, thereby preventing the system from providing proportionate, different levels of support. The system will therefore also estimate the severity of the academic risk through the use of the prediction probability score provided by the machine learning model.

The probability score, indicate how confident the model is that a particular student belongs to the Weak class; and dividing the ranges of these probability scores into sub-ranges of risk allow us to align the intensity of recommendations made to each student to their respective degrees of academic difficulties.

The mapping strategy is defined as follows:

Table 5.1: Risk Assessment Module

Weak Probability	Risk Level
<0.50	Low Risk
0.50–0.65	Mild Risk
0.65–0.80	Moderate Risk
>0.80	High Risk

This mapping allows the recommendation framework to provide different intervention intensities according to estimated academic risk severity.

3. Recommendation Module

The recommendation module generates adaptive academic support recommendations according to the assigned risk level.

A rule-based recommendation strategy is applied in this study because it provides:

- Clear intervention logic
- High interpretability
- Easy deployment
- Flexible rule expansion
- Instructor-friendly recommendation generation

The generated recommendations are intended to help instructors plan academic intervention more easily while still keeping the recommendation process transparent and understandable.

5.3. Recommendation Strategy

The recommendation module applies a rule-based intervention strategy according to the predicted risk level.

Low-Risk Students (Good): Students in this group are performing well and are not currently considered to be at risk of academic failure. The primary objective here is to sustain their existing engagement and study habits. Regular monitoring is still recommended to ensure that performance does not decline as the course progresses and assessment demands increase.

Recommended actions:

- Maintain current study performance
- Continue coursework participation
- Monitor learning progress regularly

Mild-Risk Students: At this point in time students are showing early warning signs that their performance could degrade academically unless immediate intervention occurs. A moderate Weak probability exists; while it is still considered non-critical, a small amount of self-directed effort could help avoid continued decline. Recommendations for these students will focus on providing reinforcement for basic knowledge and increasing the student's ability to be consistent in their studying to ensure that learning gaps do not develop into major obstacles.

Recommended actions:

- Increase weekly revision time
- Review previous learning materials
- Monitor coursework progress closely

Moderate-Risk Students: The Moderate Risk category is comprised of those students who have an elevated (meaningful) probability of poor academic performance. Thus, at this stage self-directed revision alone would likely be insufficient to address potential academic shortcomings. At this juncture, contact between the instructor or teaching assistant and student is necessary to identify/resolve specific areas of knowledge deficiency. Targeted practice in the area(s) where the student has demonstrated weakness will assist in solidifying the student's understanding prior to assessing the student's comprehension. Additionally, structured consultation sessions with the instructor or teaching assistant will provide the additional guidance that is required to steer the student's study habits and direction towards the most effective path possible.

Recommended actions:

- Attend consultation sessions
- Complete additional practice exercises

- Focus on weak learning topics

High-Risk Students:

At Stage 1, only 20% of course assessment data is available, yet the model has already assigned a high Weak probability to this student. This is a significant early signal that the student is struggling before major assessments have even taken place. Because little course data has been collected at this point, the intervention strategy focuses on establishing structured support as early as possible. Issuing an early warning alerts both the student and the instructor to the situation promptly. Consultation with an academic advisor helps identify whether the difficulties are academic, personal, or motivational in nature. A personalized study plan ensures that the student has a clear, achievable roadmap rather than studying without direction. Tutoring sessions provide direct academic assistance at a time when there is still sufficient opportunity to change the learning trajectory before high-stakes assessments occur.

Recommended actions:

- Early-warning intervention
- Academic advisor consultation
- Personalized study planning
- Tutoring support sessions

At Stage 2, half of the course assessment information is now available, and the model still assigns a high Weak probability. By this point, the available assessment evidence strongly confirms that the student is at serious academic risk. Because the course is already at its midpoint, the window for recovery is narrower and the interventions need to be more intensive and directly aligned with the remaining assessments. Intensive revision support addresses the accumulated knowledge gaps identified through earlier coursework performance. Remedial exercises provide structured practice targeting the specific areas where the student is weakest. Instructor consultation allows for direct academic guidance based on a more complete picture of the student's progress. Final assessment preparation support ensures that the student can still make the most of the remaining course time and maximise their performance on the assessments that carry the most weight toward the final grade.

Recommended actions include:

- Intensive revision support
- Remedial learning exercises
- Instructor consultation sessions
- Final assessment preparation support

The recommendation strategy aims to provide adaptive academic support according to estimated academic risk severity while enabling earlier intervention for academically at-risk students.

5.4. Recommendation Generation Algorithm

The recommendation generation process is implemented using a rule-based recommendation pipeline integrated with the trained machine learning prediction models.

The overall recommendation process consists of the following steps:

1. Retrieve the selected deployment model from the trained model registry
2. Input student assessment features into the prediction model
3. Generate predicted academic performance results
4. Estimate weak-class probability scores
5. Transform probability scores into academic risk levels
6. Apply predefined recommendation rules
7. Generate adaptive academic support recommendations

The rule-based approach was selected because it is:

- Interpretable
- Easy to implement
- Suitable for educational environments
- Easy for instructors to understand
- Flexible for future rule expansion

5.5. Recommendation System Implementation

The proposed recommendation framework was implemented by integrating the trained machine learning prediction models with a rule-based academic intervention mechanism. The implementation reuses the best-performing trained models obtained from the curated datasets in Chapter 4 rather than retraining new models during recommendation generation.

The overall implementation process consists of the following steps.

Step 1 — Model Selection

During the experimental phase in Chapter 4, each machine learning model was trained and evaluated under different classification settings, coursework stages, and dataset configurations. After training, the trained models were stored in a centralized model registry to allow reuse during recommendation generation without retraining the models.

The trained models were stored using a structured dictionary-based registry:

TRAINED_MODELS = {}

After each experiment, the corresponding trained model was saved using a unique configuration key containing:

- Coursework stage
- Classification setting
- Dataset type
- Model name

For example:

```
# =====  
# SAVE TRAINED MODEL  
# =====  
  
    TRAINED_MODELS[  
        ("stage1", "multi", "curated", model_name)  
    ] = results["Trained_Model"]
```

This registry structure allows the recommendation framework to retrieve different trained models according to the selected educational monitoring scenario.

To avoid object reference overwriting during model storage, deep-copy operations were applied when saving trained models into the registry.

Step 2 —Deployment Model Selection

After model selection, the recommendation framework automatically selected the best-performing deployment models according to:

- Weak-student detection capability
- Overall prediction performance
- Model stability

The selected deployment models were then stored within the recommendation framework configuration:

```
RECOMMENDATION_MODELS = {  
  
    "20%": TRAINED_MODELS[  
        (  
            "stage1",  
            "binary",  
            "curated",  
            best_stage1["Model"]  
        )  
    ],  
  
    "50%": TRAINED_MODELS[  
        (  
            "stage2",  
            "binary",  
            "curated",  
            best_stage2["Model"]  
        )  
    ]  
}
```

Step 3 — Student Performance Prediction

During recommendation generation, student assessment data is provided as input to the deployed prediction models.

The selected model predicts:

- Predicted academic class
- Weak-class probability score

The prediction probability is subsequently used for probability-based academic risk estimation.

Step 4 — Recommendation Output Generation

Finally, the framework generates structured recommendation outputs containing:

- Student identifier
- Predicted academic performance
- Academic risk level
- Generated academic support recommendations

The final output is presented in tabular form to facilitate educational monitoring and intervention planning.

5.6. Example Recommendation Output

To demonstrate the functionality of the proposed recommendation framework, example outputs from the deployed recommendation scenarios are presented below.

Scenario 1 —Early Warning

The Stage 1 binary configuration focuses on early identification of at-risk students using limited coursework information collected during the early stage of the course.

Table 5.2: Early Warning example output

Student_ID	Real_Class	Predicted_Class	Risk_Level	Recommendation
student000000	G	Good	Low Risk	Maintain current study performance, Continue coursework participation
student000001	G	Good	Low Risk	Maintain current study performance, Continue coursework participation
student000002	G	Weak	Mild Risk	Increase weekly revision time, Review previous learning materials
student000003	G	Good	Low Risk	Maintain current study performance, Continue coursework participation

The results demonstrate that early-stage prediction can identify some academically weak students before major assessments occur. However, prediction errors may still occur because limited learning evidence is available during the early stage of the course.

Scenario 2 — Mid-course Monitoring

The Stage 2 binary configuration utilizes additional assessment information collected during the middle stage of the course, resulting in more reliable academic risk prediction.

Table 5.3: Mid-course Monitoring example output

Student_ID	Real_Class	Predicted_Class	Risk_Level	Recommendation
student000081	W	Weak	High Risk	Receive intensive revision support, Complete remedial learning exercises
student000089	W	Weak	High Risk	Receive intensive revision support, Complete remedial learning exercises
student000092	W	Weak	High Risk	Receive intensive revision support, Complete remedial learning exercises
student000100	W	Weak	High Risk	Receive intensive revision support, Complete remedial learning exercises

Compared with the Stage 1 configuration, the Stage 2 recommendation framework produces more stable weak-student detection and more reliable academic intervention recommendations due to the availability of additional coursework information collected during later stages of the course.

5.7. Educational Implications

The proposed recommendation system provides several practical implications for educational environments.

First, the framework supports earlier identification of academically at-risk students, allowing instructors to provide timely intervention before learning difficulties become more severe. Early identification may help improve student learning outcomes and reduce the risk of academic failure during later stages of the course.

Second, the recommendation process generates adaptive academic support suggestions according to predicted student risk levels. This allows students to receive more appropriate learning guidance based on their estimated academic condition and intervention needs.

Third, the recommendation mechanism remains simple and interpretable due to the use of rule-based intervention strategies. Instructors can easily understand how recommendations are generated and may further modify the recommendation rules according to different educational contexts or institutional requirements.

Finally, the proposed system may help reduce instructor workload by automatically identifying students who may require additional academic support and generating preliminary intervention recommendations for educational monitoring purposes.

5.8. Limitations of the Recommendation System

Several limitations of the proposed recommendation framework should be acknowledged.

The current recommendation mechanism relies primarily on manually designed rule-based intervention strategies. Although this approach improves interpretability and implementation simplicity, more advanced recommendation techniques such as collaborative filtering, knowledge-based recommendation, or reinforcement learning may provide more adaptive and personalized recommendation generation in future studies.

Additionally, the probability thresholds map prediction scores into discrete risk categories were determined heuristically to facilitate practical deployment. Although these cut-offs offer a clear way to vary intervention intensity, they may not be optimal across all educational settings. Future research could explore data-driven threshold optimization techniques or incorporate expert validation to improve the reliability and generalizability of the risk assessment process.

Furthermore, the quality of generated recommendations is closely related to the prediction performance of the deployed machine learning models. Incorrect performance predictions may lead to unsuitable academic intervention recommendations, particularly for students located near the decision boundary between academic risk categories.

The current framework also mainly utilizes assessment-related academic features during prediction and recommendation generation. Additional information such as learning behavior, LMS interaction patterns, psychological factors, or demographic attributes may further improve recommendation quality and personalization capability.

Finally, the recommendation framework has not yet been evaluated in real classroom environments or integrated into actual educational platforms. Future work may therefore focus on real-world deployment and evaluation to better measure the effectiveness of the generated intervention strategies on student learning outcomes.

CHAPTER 6

CONCLUSION AND FUTURE WORK

6.1. Conclusion

This study explored several machine learning models for early student performance prediction using both binary and multi-class classification settings. The experiments were carried out at two different stages of the semester, using assessment data from roughly 20% and 50% of the coursework.

In general, the models run in Stage 2 outperformed those run in Stage 1; and binary classification was much more consistent than multi-classification.

The models themselves also behaved a bit differently. Random Forest usually gave the strongest overall performance in the binary experiments. On the other hand, SVM worked quite well in several Stage 2 experiments that focused on weak-student detection. Logistic Regression gave stable results overall, although its scores were often lower than Random Forest.

Multi-class classification turned out to be more difficult than expected, especially during Stage 1. This was especially obvious during Stage 1 because the amount of available data was still limited. Some minority student groups were difficult for the models to classify consistently under the imbalanced dataset setup used in this study.

This study also experimented with BorderlineSMOTE and epiplexity-guided synthetic sample filtering to deal with the imbalance problem. In some experiments, the curated synthetic dataset helped improve the results. In other cases, the improvement was smaller or not very noticeable depending on the model and prediction stage.

Besides the prediction models, a simple rule-based recommendation component was also developed. The recommendations were generated using the predicted student performance levels and were mainly intended to provide different forms of academic support for students with different risk levels.

To sum up, early student performance prediction is still challenging when only a small amount of coursework information is available and the dataset remains highly imbalanced. Even so, the experiments showed that machine learning models, together with curated synthetic sampling, can still help identify students who may be academically at risk earlier in the semester.

6.2. Future work

Several ways exist for future research on improving the proposed framework.

- Large-scale educational databases with a variety of student populations and course types may be used by researchers in the future to obtain better results that generalize across many learning environments.
- Future research may use other student related information in addition to academic achievement (e.g., attendance, LMS activity, demographics, psychological factors) to generate predictions that are even stronger when predicting students at an early time in their development.
- Two possible avenues of exploration for addressing the problem of class distribution imbalance mentioned above have already been discussed briefly; however, there was no consistency in which imbalance methods resulted in improved results across all models and all stages of prediction. It is thus worthwhile to investigate other imbalance method(s) or alternative synthetic sampling techniques.
- Also worthy of investigation in future studies would be more sophisticated machine learning methods than those currently being employed, such as deep learning models, hybrid approaches, or ensemble learning. These may produce superior results given the complexity of many educational databases.
- The recommendation system could potentially be expanded into an adaptive or intelligent recommendation system through the inclusion of collaborative filtering, reinforcement learning, or Explainable Artificial Intelligence (XAI).
- It is important for future research to test the usefulness of the recommendations produced by the framework in actual education settings.

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